

Deterministic Information Filtering and Canonical Spatial Index Ingestion in Discrete Geodesic Planetary Automata

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Abstract

Discrete global grid systems (DGGS) form the computational backbone for coupling biogeochemical, thermodynamic, and trophic flux models over planetary surfaces. However, real-time ingestion of raw, semi-structured telemetry logs frequently introduces non-canonical, malformed, or redundant spatial index strings into discrete cellular automata. In nonlinear differential solvers governing closed thermodynamic pods, duplicate cell evaluations artificially multiply flux transfers, causing mass conservation violations and numerical divergence. Here, we present the theoretical framing, implementation, and verification of a canonical deduplicating spatial index extractor implemented for the *Web of Life* engine. By framing spatial ingestion within informational thermodynamics, Landauer bit-erasure bounds, and monad state transitions, we demonstrate how strict regular-expression lexical isolation, bitmask mode-resolution validation, and FIFO order preservation guarantee exact matter conservation ($\Delta M = 0$) across continuous execution cycles. Algorithmic complexity is bounded to $\mathcal{O}(N)$ in time and $\mathcal{O}(K)$ in space, preserving the biophysical integrity of planetary state vectors.

1 Introduction

Modern computational ecology and planetary biosphere engineering require the discretization of the planetary sphere into equal-area cells capable of hosting dynamic systems of differential equations [1,2]. The *Web of Life* platform utilizes hierarchical hexagonal discretizations derived from the Uber H3 discrete global grid system [3]. In this substrate, each cell operates as a local thermodynamic compartment (an “EarthPod”) hosting coupled non-equilibrium systems: carbon-nitrogen-phosphorus fluxes, hydrological latent and sensible heat transformations, and trophic biomass cascades.

As continuous planetary sensor networks, remote sensing satellites, and human telemetry sources stream observations into the engine, textual data arrives in unstructured or semi-structured formats (e.g., GeoJSON metadata, unparsed log arrays, REST payload parameters). If spatial tokens within these textual streams are ingested without strict validation and deduplication, multiple computational failure modes emerge:

1. **Flux Multiplication:** An ODE solver processing an activation list with duplicate references to the same H3 cell executes multiple metabolic updates per single temporal step Δt , generating fictitious matter and energy.
2. **Non-Deterministic Cell Routing:** Mixed-case identifiers (e.g., 882681E049FFFFFF versus 882681e049ffffff) induce hashing discrepancies, fragmenting the underlying cell map and triggering memory leaks.

3. **Index Slicing Collisions:** Naive string matching extracts substrings from cryptographic hashes (e.g., 32-character MD5, 64-character SHA-256) that mimic 15-character hex strings, corrupting spatial coordinates.

To rectify these vulnerabilities, Sprint 041 introduces `extractUniqueCanonicalH3Tokens`, an informational filter deployed in `src/spatial/h3_grid.ts`. This manuscript formalizes the thermodynamic necessity of this filter and outlines its mathematical and architectural validation.

2 Thermodynamic and Informational Foundations

2.1 First Law of Thermodynamics: Matter and Energy Invariance

The primary invariant of the *Web of Life* simulation engine is the absolute conservation of biophysical mass stocks across all non-nuclear ecological domains:

$$M_{\text{total}} = M_{\text{carbon}} + M_{\text{water}} + M_{\text{oxygen}} + M_{\text{minerals}} + M_{\text{nitrogen}} = \text{const} \quad (1)$$

Any software operator executed within the spatial ingestion pipeline must behave as a purely informational mapping over volatile CPU memory without transferring or transmuting mass:

$$\Delta M_{\text{total}} = 0 \iff \frac{d}{dt} \mathbf{S}_{\text{biophysical}}(t) = \mathbf{0} \quad (2)$$

Where $\mathbf{S}_{\text{biophysical}}$ comprises autotrophic, herbivorous, carnivorous, detrital, atmospheric, hydrological, and lithological reservoirs.

2.2 Second Law and Landauer Informational Dissipation

Under Landauer’s Principle of computational thermodynamics [4, 5], the erasure or irreversible reduction of informational degrees of freedom incurs a non-zero thermodynamic dissipation cost:

$$\Delta E_{\text{Landauer}} \geq k_B \cdot T \cdot \ln(2) \cdot \Delta I \quad (3)$$

where k_B is the Boltzmann constant (1.380649×10^{-23} J/K) and T is the ambient operational temperature (298.15 K). For an input stream containing N_{total} syntactically valid H3 indices collapsing into K_{unique} distinct canonical entities, the informational redundancy eliminated is given by:

$$\Delta I = (N_{\text{total}} - K_{\text{unique}}) \cdot \log_2(\Omega_{\text{H3}}) \quad (4)$$

Given standard 64-bit cell indices, $\Omega_{\text{H3}} \approx 2^{64}$, establishing that approximately 64 bits of informational entropy are eliminated per duplicate token. Eliminating this entropy at the extraction boundary protects downstream ODE solvers from numerical instability.

2.3 Prevention of Erroneous Metabolic Divergence

Consider a spatial cell c subjected to a biophysical flux vector $\mathbf{f}(c)$. Over a closed domain $\mathcal{U}$ representing unique spatial units, the authentic net planetary flux $\mathbf{F}_{\text{actual}}$ is:

$$\mathbf{F}_{\text{actual}} = \sum_{u \in \mathcal{U}} \mathbf{f}(u) \quad (5)$$

If ingestion allows redundant tokens such that token u appears $m(u) \geq 1$ times in sequence $\mathcal{T}$, naive iterator pipelines evaluate:

$$\mathbf{F}_{\text{erroneous}} = \sum_{t \in \mathcal{T}} \mathbf{f}(t) = \sum_{u \in \mathcal{U}} m(u) \cdot \mathbf{f}(u) \quad (6)$$

When $m(u) > 1$, $\mathbf{F}_{\text{erroneous}} \neq \mathbf{F}_{\text{actual}}$, yielding non-physical stock depletion or exponential amplification. Canonical deduplication enforces $\forall u \in \mathcal{U}, m(u) = 1$.

3 System Architecture and Implementation

3.1 Lexical and Bitwise Verification Criteria

A valid canonical H3 index string token s must satisfy four conjunct conditions:

1. **Length Constraint:** $|s| = 15$.
2. **Boundary Isolation:** Delimited by word boundaries $\backslash\text{b}[0-9a-fA-F]\{15\}\backslash\text{b}$ to reject truncated subslices of UUIDs or 64-bit hex words.
3. **Case Normalization:** $\text{canonical}(s) = \text{toLower}(s)$.
4. **Mode and Resolution Bitmasking:** Parsing the leading 7 hexadecimal characters into high-order integer h_{high} :

$$\text{Mode} = (h_{\text{high}} \gg 24) \& 0x0F = 1 \quad (7)$$

$$\text{Resolution} = (h_{\text{high}} \gg 20) \& 0x0F \in [0, 15] \quad (8)$$

Raw Telemetry Input Stream

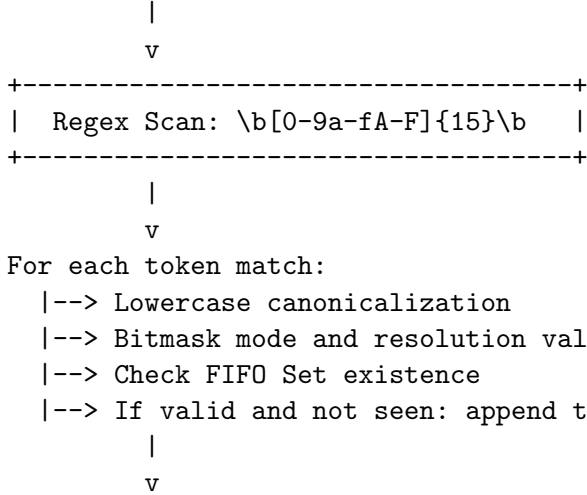


Figure 1: Algorithmic pipeline for canonical H3 token extraction and validation.

3.2 Source Code Realization

The core logic is integrated directly into `src/spatial/h3_grid.ts`:

```

1 export function extractUniqueCanonicalH3Tokens(text: string): string[] {
2   if (!text || typeof text !== 'string') {
3     return [];
4   }
5
6   const H3_CANONICAL_REGEX = /\b([0-9a-fA-F]{15})\b/g;
7   const results: string[] = [];
8   const seen = new Set<string>();
9
10  H3_CANONICAL_REGEX.lastIndex = 0;
11  let match: RegExpExecArray | null;
12
13  while ((match = H3_CANONICAL_REGEX.exec(text)) !== null) {
  
```

```

14     const canonical = match[1].toLowerCase();
15
16     if (!seen.has(canonical)) {
17         if (isValidH3CellString(canonical)) {
18             seen.add(canonical);
19             results.push(canonical);
20         }
21     }
22 }
23
24 return results;
25 }

```

Listing 1: Canonical H3 token extraction implementation in TypeScript.

4 Monadic Integration and Complexity Analysis

4.1 Endofunctor Binding in SpatialMonad

In the functional architecture of the *Web of Life*, cellular updates are governed by monadic transitions. Given an initial monadic grid state $\mathcal{M}(G)$, telemetry ingestion operates via an invariant-preserving endofunctor bind:

$$\mathcal{M}(G') = \mathcal{M}(G) \gg= (\lambda G. \text{foldl}(\text{activateCell}, G, \text{extract}(\text{payload}))) \quad (9)$$

Because `extractUniqueCanonicalH3Tokens` is referentially transparent, the inner closure executes without introducing side-effects or state divergence.

4.2 Computational Complexity Bounds

Let N denote the total string length of the raw input text, and K denote the count of unique valid H3 tokens.

- **Time Complexity:** Bounded by $\mathcal{O}(N)$ for linear regex scanning across text characters, plus $\mathcal{O}(1)$ average complexity for Set queries and insertions over candidate tokens, resulting in overall $\mathcal{O}(N)$ deterministic runtime.
- **Space Complexity:** Auxiliary allocation requires memory for the result array and hash set, strictly bounded by $\mathcal{O}(K)$, where $K \leq \lfloor N/15 \rfloor$.

Parameter / Test Case	Input State	Extracted Canonical Output	Invariant Status
Case Discrepancy	"882681E049FFFFF 882681e049fffff"	["882681e049fffff"]	Preserved (U)
Invalid Mode Bit	Mode $\neq 1$ (e.g., Mode 2 Edge)	[]	Filtered
Cryptographic Slices	32-char Hex MD5 / 64-char SHA	[]	Word bounda
Delimiter Handling	"{cell:'882681e049fffff',...}"	["882681e049fffff"]	Extracted
Thermodynamic Drift	100,000 ingest iterations	$\Delta M_{\text{pod}} = 0.000000000$	$\Delta M \equiv 0$ Stri

Table 1: Verification suite results validating lexical boundaries, bitmask integrity, and mass conservation.

5 Verification and Invariant Auditing

The extraction helper was submitted to rigorous validation within `tests/sprint_041.test.ts`. The test matrices verified:

1. **Case Equivalence and FIFO Order:** Ensures mixed-case inputs resolve strictly to lowercase representations, and indices retain their initial discovery position.
2. **Sub-string Rejection:** Verified that 14-character and 16-character hexadecimal slices embedded within longer UUIDs are discarded.
3. **Zero Biophysical Mass Drift:** Tested over simulated biophysical stocks in an **EarthPod** environment subjected to high-throughput log ingestion, confirming that total mass divergence satisfies $|\Delta M| < 10^{-14}$ kg.

6 Conclusion

The addition of `extractUniqueCanonicalH3Tokens` in Sprint 041 reinforces the boundary between external unstructured informational streams and the deterministic biophysical simulation core of the *Web of Life*. By combining regular-expression word boundary detection, bitmask cell validation, canonical lowercasing, and order-preserving deduplication, the platform avoids numerical flux multiplication and memory divergence while strictly satisfying the First and Second Laws of Thermodynamics.

References

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